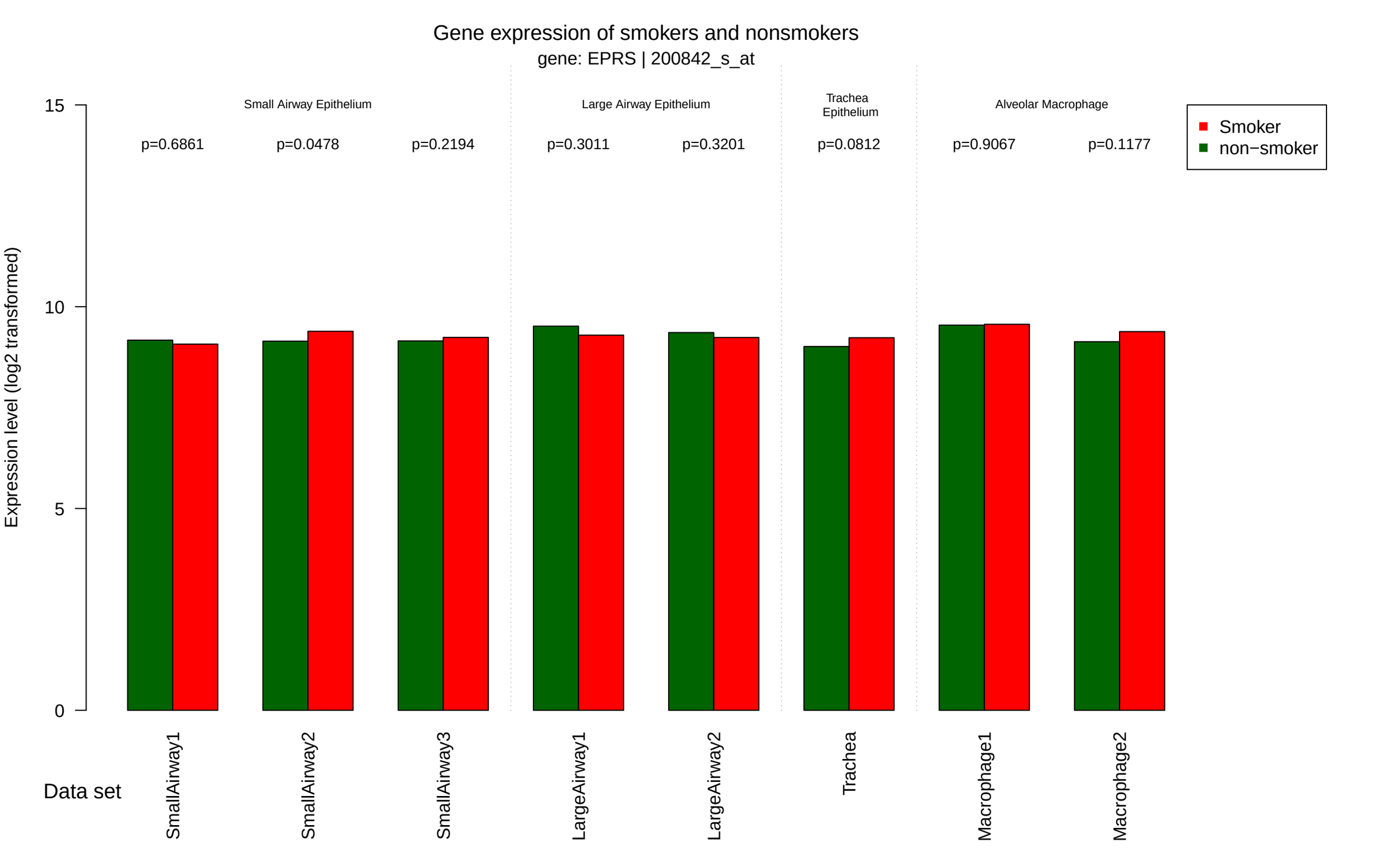
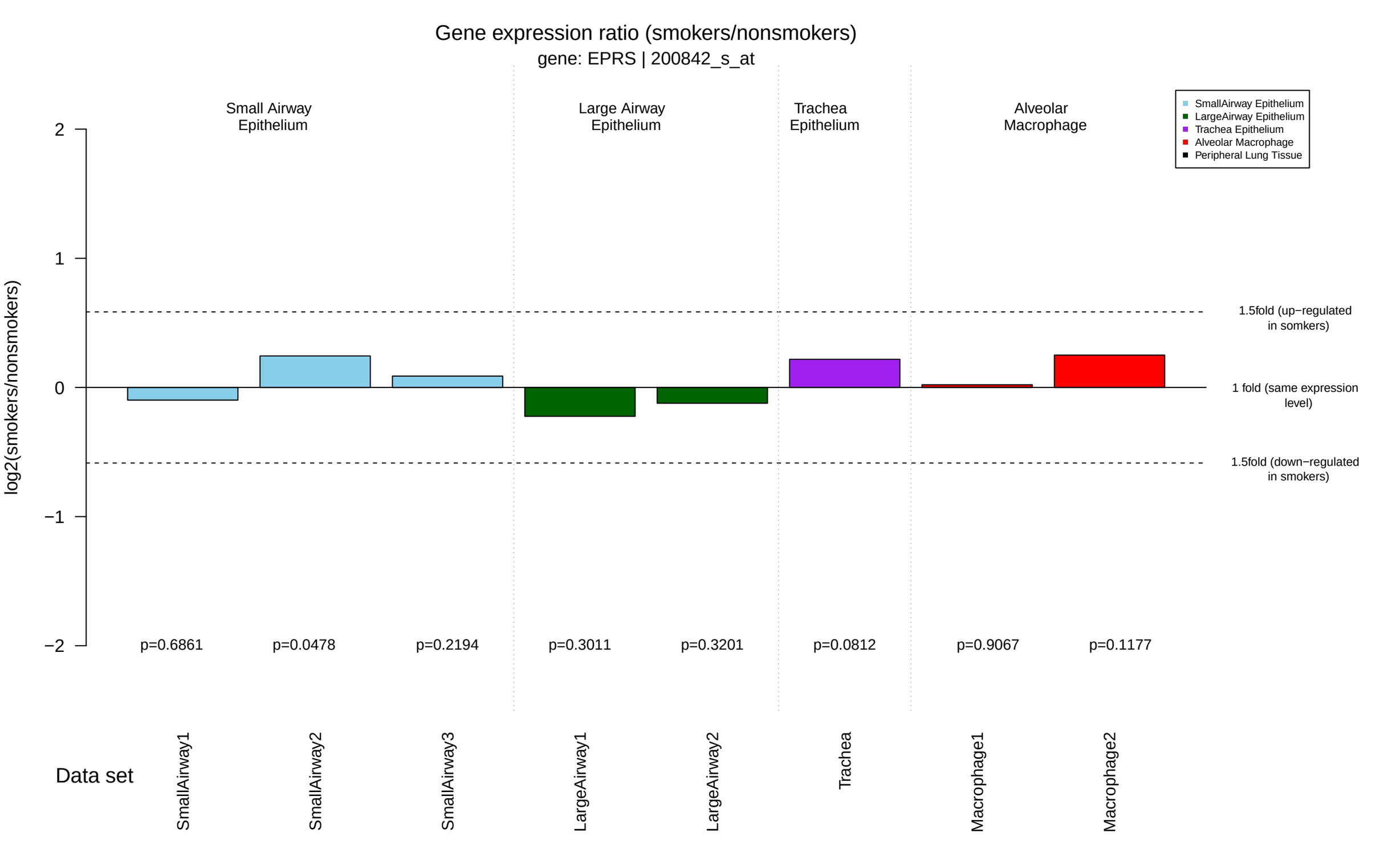
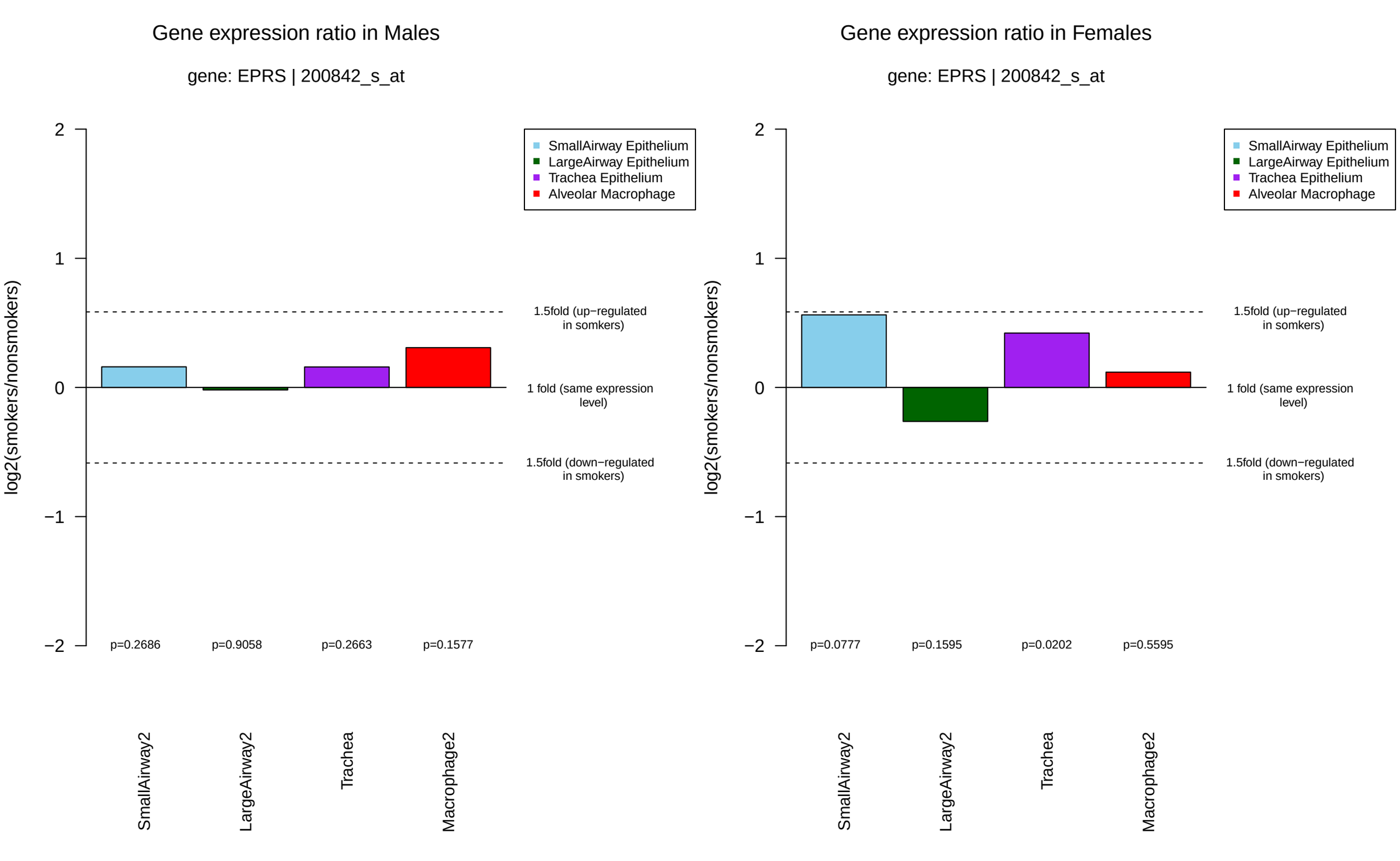
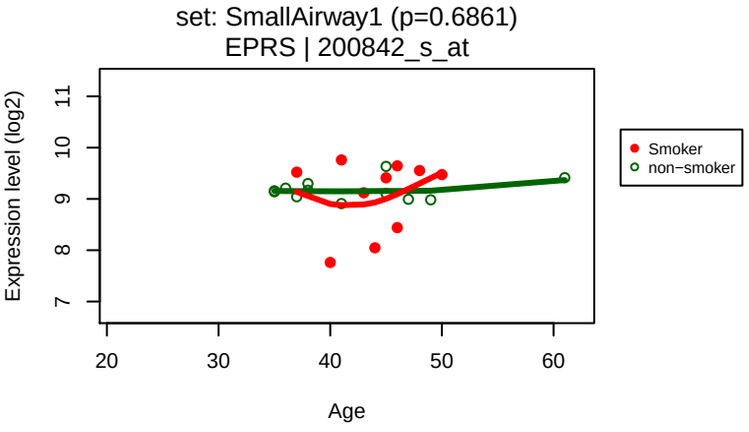
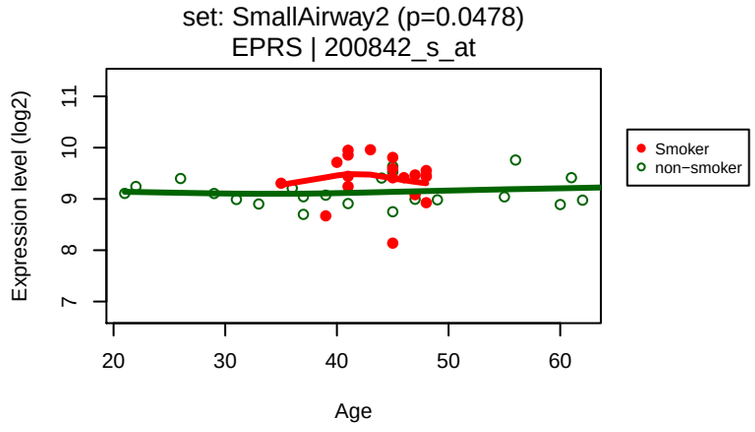
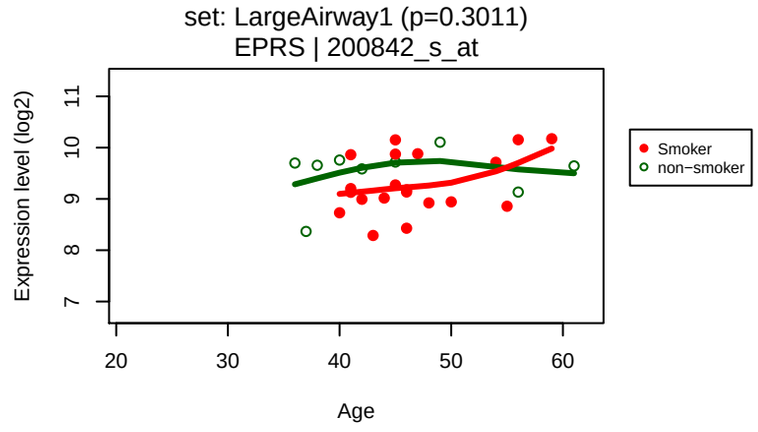

Gene expression of smokers and nonsmokers
gene: EPRS | 200842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6861
p=0.0478
p=0.2194
p=0.3011
p=0.3201
p=0.0812
p=0.9067
p=0.1177
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

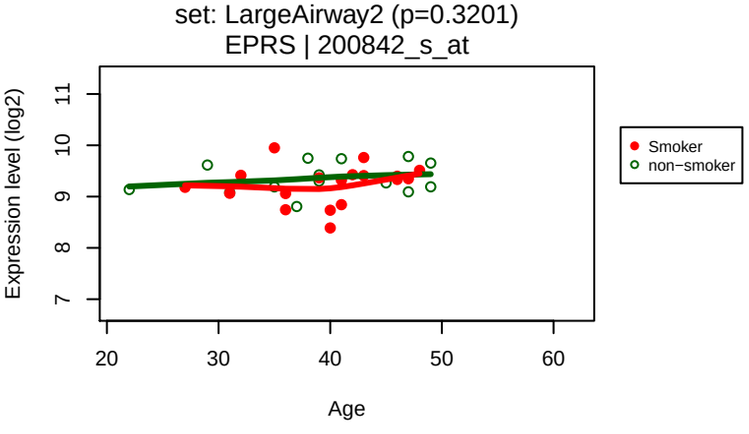

Gene expression ratio (smokers/nonsmokers)
gene: EPRS | 200842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6861
p=0.0478
p=0.2194
p=0.3011
p=0.3201
p=0.0812
p=0.9067
p=0.1177
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

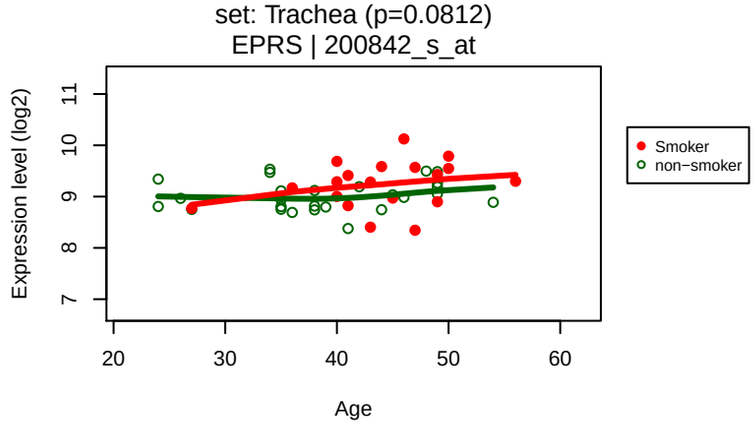

Gene expression ratio in Males
gene: EPRS | 200842_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2686
p=0.9058
p=0.2663
p=0.1577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EPRS | 200842_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0777
p=0.1595
p=0.0202
p=0.5595
SmallAirway2
LargeAirway2
Trachea
Macrophage2

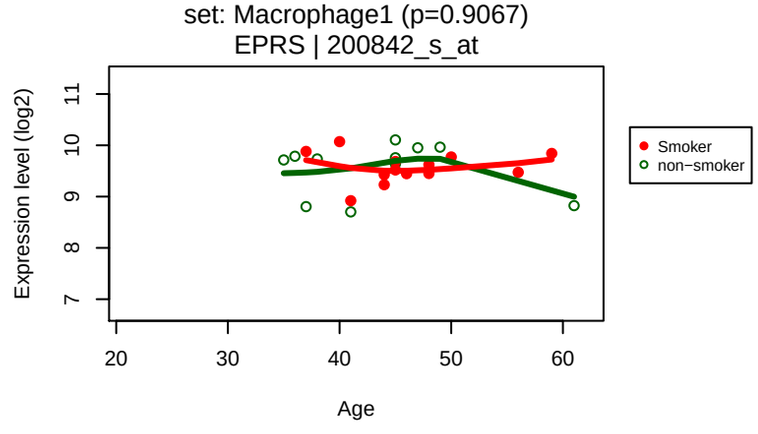

set: SmallAirway1 (p=0.6861)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker

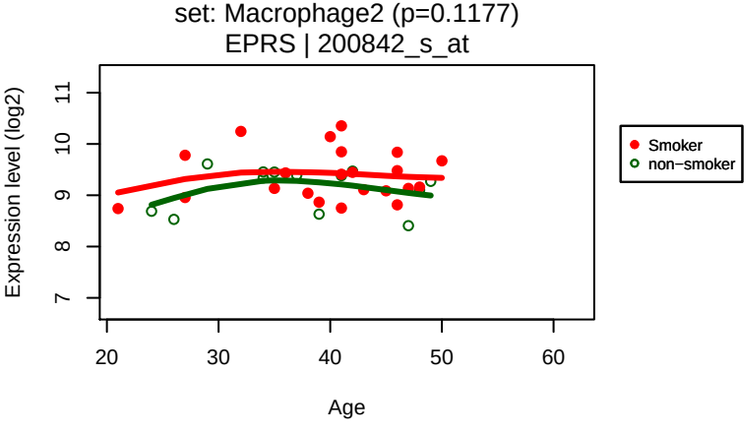

set: SmallAirway2 (p=0.0478)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker

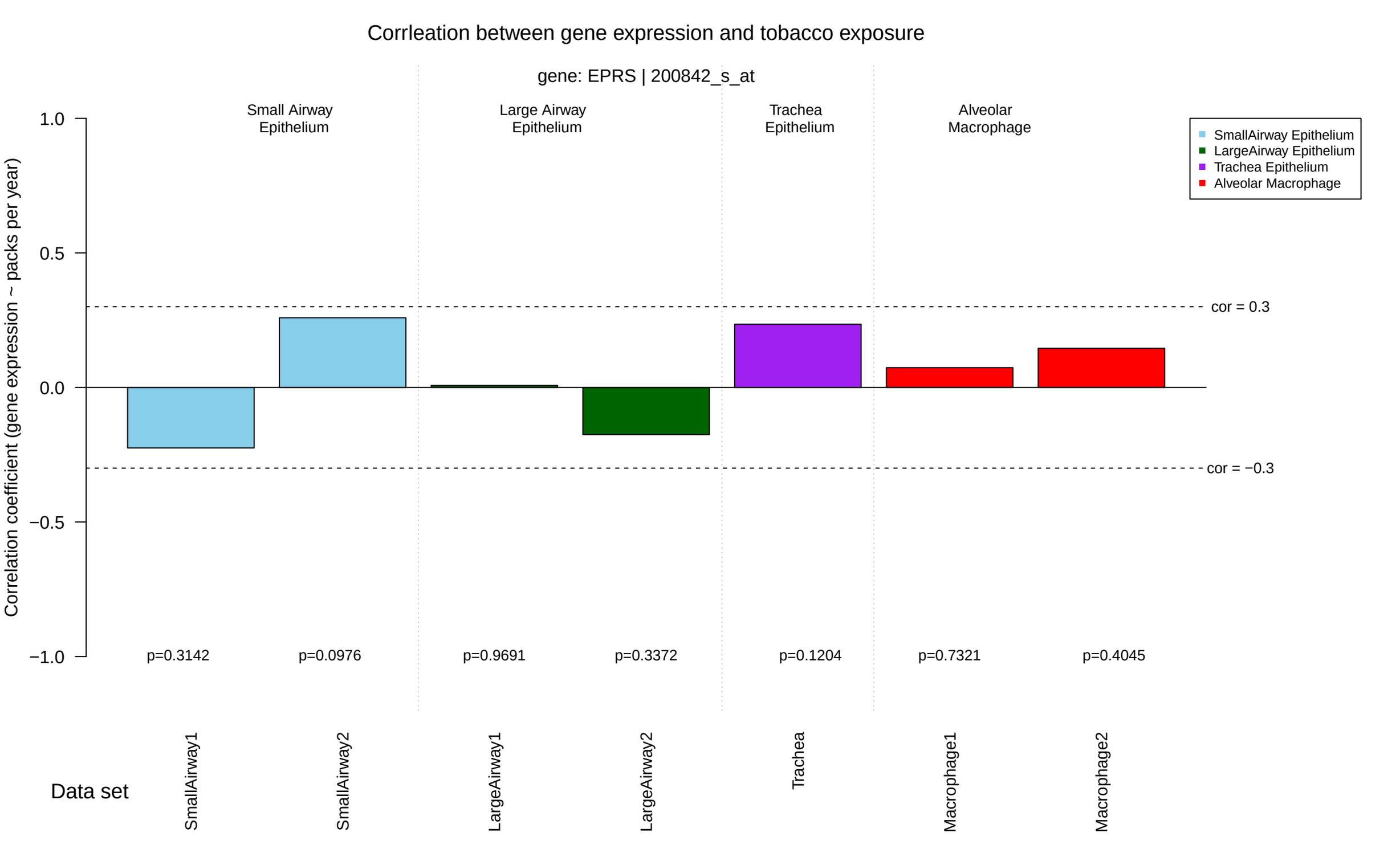

set: LargeAirway1 (p=0.3011)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3201)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0812)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9067)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1177)
EPRS | 200842_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EPRS | 200842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3142
p=0.0976
p=0.9691
p=0.3372
p=0.1204
p=0.7321
p=0.4045
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2